Author2vec: A Jacobian Lens for Authorship Identity in Embedding Encoders

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Abstract

Anthropic's *"Verbalizable Representations Form a Global Workspace in Language Models"* introduces the averaged-Jacobian lens (J-lens) and argues that a large closed decoder (Claude Sonnet 4.5, 127 layers) maintains a privileged "workspace" of verbalizable, causally-active concepts. We adapt that apparatus to a setting it was not designed for, small open embedding encoders whose output is a single masked-mean-pooled vector, and re-point it from reasoning to personal writing style, or authorship identity. Three ingredients make the transplant work: a δ-broadcast reduction that collapses the encoder's position×position Jacobian to one matrix per layer; an embedding-space projection that removes the meaningless radial direction of a normalized embedding; and a style lens that reads the layer Jacobian onto interpretable identity axes rather than the token vocabulary. On a 6-layer prose encoder (MiniLM) and a 12-layer code encoder (JinaBERT), author identity is a computed intermediate, decodable above chance at every layer and not only at the output. The identity direction serves as both a detector (is a person's fingerprint in the weights?) and a causal lever (steering), which we probe further with an ignition experiment and on a generative decoder. Validated against a measured population, the same embeddings recover self-reported Big Five personality from prose (Openness +6.8 points over the majority baseline; all five traits beat a shuffled-label null at $p < 0.001$), while an astrological-sign negative control on a separate blog corpus recovers nothing ($p = 0.96$). The personality signal is only a few points above chance, which is also why we make no cognitive-capacity claim. Everything runs on open models, ships as static assets, and reproduces on a laptop. We separate replication from new results throughout, and claim no "IQ" and no consciousness.

Contributions

1. **An encoder adaptation of the averaged-Jacobian J-lens** via the δ-broadcast reduction (§4.2), redefining the method for a masked-mean-pooled, non-generative encoder.
2. **An embedding-space Jacobian projection** for L2-normalized outputs (§4.3), unit-tested orthogonal to the embedding.
3. **The style lens** (§4.5): a Jacobian readout onto interpretable identity axes rather than the token vocabulary, the trustworthy signal where an encoder has no clean unembedding.
4. **Identity is a computed intermediate** (§5.1): author identity decodes above chance from the internal Jacobian at every layer, and the best internal code layer matches the output-embedding ceiling.
5. **A fingerprint-presence detector** (§5.4): known identity vs. "blank space" for out-of-distribution text, a question the reference paper does not ask.
6. **Identity ignition** (§5.2): the representation commits to a single individual increasingly with depth, peaking mid-network for prose and late for code, above two null controls.
7. **Causal steering and directed modulation** (§5.4, §5.6): the identity direction is a sign-controllable lever on encoders, and a leakage-free concept direction is a modest but real causal handle on a decoder's output.

8. **A measured-population validation with a negative control** (§5.7): on 2,467 psychometrically-labelled essays the same embeddings recover self-reported Big Five personality above a shuffled-label null on all five traits (mean lift +4.6 points over majority; Openness +6.8; all five $p < 0.001$ by a 1000-permutation test). On the Blog Authorship Corpus, writer age is recovered ($p < 0.001$) while an astrological-sign negative control is not ($p = 0.96$).
9. **An open, reproducible, browser-native reimplementation** across two modalities, prose and code, with a faithfulness gate and a full audit ledger.

The new method is narrow: the δ-broadcast reduction (1) and the embedding-space projection (2), which make the averaged-Jacobian lens run on a pooled encoder at all. Contributions 3–7 apply established techniques (difference-of-means directions, linear probing across depth, the ignition design) to a new target, authorship identity read through the layer Jacobian; 8–9 are the measured-population validation and the open reimplementation.

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1. Introduction

Recent interpretability work argues that large language models maintain a privileged, verbalizable "workspace" of representations: a small subset of activations that are reportable, controllable, and causally responsible for reasoning (Anthropic 2026). That evidence comes from a generative decoder with hundreds of billions of parameters and privileged internal access, read out with an averaged-Jacobian lens onto the token vocabulary. It is expensive to reproduce and impossible to inspect from the outside.

We ask a smaller, checkable question with the same apparatus: where, inside a network, does *who wrote this* become decidable? We move the averaged-Jacobian lens from a frontier decoder to two small, open embedding encoders, a 6-layer prose model (MiniLM) and a 12-layer code model (JinaBERT), and re-point it from reasoning onto personal writing-style identity. An encoder is a minimal testbed for the reference paper's mechanistic claims because it strips away generation: there is no autoregressive loop to smuggle information through, only a fixed map from text to a single pooled vector. It cannot speak to the behavioral claims (verbal report, reasoning swaps); those need a decoder, which we take up separately (§5.5–5.7).

Carrying the lens across that gap takes three ingredients (§4): a δ-broadcast reduction that collapses the encoder's intractable position×position Jacobian to one matrix per layer; an embedding-space projection that removes the meaningless radial direction of a normalized output; and a style lens that reads the layer Jacobian onto interpretable identity axes rather than the token vocabulary, the trustworthy signal where an encoder has no clean unembedding.

With these we find that author identity is a computed intermediate, decodable above chance at every layer (§5.1) and not only at the output embedding; that an ambiguous two-author input drives the internal representation to commit to a single author increasingly with depth, peaking at a characteristic depth (mid-network for prose, late for code) and surviving two null controls (§5.2); and that the same geometry yields a fingerprint-presence detector separating a known identity from "blank space" (§5.4). These signatures hold across both modalities. We keep the replication-versus-new boundary sharp throughout and are explicit about what we do not claim (§7); every result number is machine-extracted into the audit ledger (Appendix D) and gated by the review harness.

2. Related work

Lenses on the residual stream. The logit lens (nostalgebraist 2020) reads intermediate activations through the unembedding; the tuned lens (Belrose et al. 2023) learns an affine correction per layer. The averaged Jacobian of (Anthropic 2026) generalizes this by linearizing the *whole* map from a layer to the output and averaging over a corpus. We inherit that construction and adapt it to a pooled encoder (§4.2); our vocab lens is a logit lens over tied WordPiece embeddings, and our *style lens* (§4.5) replaces the vocabulary readout with one onto interpretable directions, the piece that is new here.

Steering and concept directions. Activation addition (Turner et al. 2023) and contrastive activation addition (Rimsky et al. 2024) steer generation by adding a direction to the residual stream; representation engineering (Zou et al. 2023) extracts such directions, often as a difference of class means. A single such direction can be a causal lever (as in refusal, mediated by one direction (Arditi et al. 2024)), and causal-mediation analysis localizes where a direction acts (Vig et al. 2020). Our identity axes are difference-of-means (Fisher) directions, used both to *read* (§4.5) and to *steer* (§5.4); the novelty is not the technique but the target, personal authorship identity, and reading it *through the layer Jacobian*.

Probing and representational geometry. Linear probes (Alain and Bengio 2017; Hewitt and Manning 2019) test what a layer linearly encodes; because above-chance probing need not imply the model *uses* the information (Hewitt and Liang 2019), we pair every decode with control-direction and shuffled-label nulls. We use leave-one-out nearest-centroid decoding as a probe of *identity* across depth (§5.1). Linear CKA (Kornblith et al. 2019) compares representations between layers, which we apply to J-lens readout geometry (§4.7). Sparse dictionary learning / SAEs (Bricken et al. 2023) decompose activations into interpretable atoms; the paper's J-space decomposition is a supervised cousin we reuse (§4.6).

Global workspace and authorship. The framing of (Anthropic 2026) draws on global workspace theory (Baars 1988) and the "ignition" of conscious access (Dehaene and Changeux 2011); we borrow the *ignition* experimental design (§5.2) but point it at identity, and we make no claim about consciousness. Our substrate is Sentence-BERT-style embeddings (Reimers and Gurevych 2019), and the downstream question, attributing text to its author from style, is classical authorship attribution / stylometry (Mosteller and Wallace 1964; Stamatatos 2009), now often approached with learned authorship embeddings (Rivera-Soto et al. 2021), here recast as a question about *where in a network* identity is computed. A known hazard is the content–style confound: apparent authorship signal can be topic signal (Wegmann et al. 2022), which we control on the model side (§5.10) and flag as a limitation on the prose side (§7). Throughout, the J-lens, J-space, and structural signatures are the paper's; our contribution is their transplant to open encoders, the style-lens readout, the identity target, and full reproducibility.

3. Background: the averaged-Jacobian lens

We build directly on the apparatus of (Anthropic 2026), which we summarize here so that our adaptation (§4) and its boundaries are unambiguous. Quantitative figures in this section (layer counts, workspace band, J-space fractions) are that paper's *reported* values; it is a single online article, so we attribute them to it as a whole rather than by page.

The J-lens. For a causal decoder, the *averaged Jacobian* at layer ℓ is

$$J_\ell \;=\; \mathbb{E}_{t,\,t'\ge t,\,\mathrm{prompt}}\left[\frac{\partial h_{\mathrm{final},t'}}{\partial h_{\ell,t}}\right],$$

the linearized effect of a layer-ℓ activation on the final residual stream, averaged over token positions and a corpus of prompts. Read through the unembedding it yields, for any activation, a ranked list of vocabulary tokens the model is "disposed to say", a corrected logit lens that accounts for representational change across layers.

J-space and its privilege. Activations are decomposed into sparse non-negative combinations of k ($\approx$ 10–25) J-lens vectors; the *J-space component* is the part of an activation lying in that cone. Though it accounts for only $\sim$ 6–10% of a concept vector's variance, it is the part causally responsible for the concept's availability to verbal report, and it is subject to top-down control (instructing the model to "focus on X" loads X), mediates unverbalized reasoning intermediates (swapping a J-lens vector redirects downstream answers), and is required for deliberate (but not automatic) tasks.

Structural depth signatures. Four quantities, read off J_ℓ across depth (the paper's Figure 28), identify a workspace band: next-token-prediction accuracy of the readout, its *excess kurtosis* (peakiness), the *autocorrelation* of the top lens token across positions (persistence of abstract content), and the readout's *effective linear dimensionality.* All four mark a consistent sensory $\to$ workspace $\to$ motor tripartition (noisy early layers, a coherent middle band, and output-aligned late layers), corroborated by an ambiguous-input "ignition" experiment in which the representation snaps to one interpretation at workspace onset. The study is conducted on Claude Sonnet 4.5 (127 layers; workspace ≈ L38–92) and corroborated on Haiku/Opus.

The gap we step into. Every one of these constructs is defined for a *generative decoder* with per-position next-token logits. An embedding encoder has neither: its output is a single pooled vector. §4 is what it takes to carry the mechanistic half of this apparatus across that gap; §5.2 and §5.5–5.7 are our attempts at the behavioral half the encoder cannot reach.

4. Method

Setup. We study two open embedding encoders: `sentence-transformers/all-MiniLM-L6-v2` (Wang et al. 2020) (prose; 6 layers, $d = 384$, vocab 30,522) and `jinaai/jina-embeddings-v2-base-code` (Günther et al. 2023) (code; 12 layers, $d = 768$, vocab 61,056). Each maps a passage to one masked-mean-pooled, L2-normalized vector p. A faithfulness gate (`bin/spike`) asserts our native candle forward reproduces the shipped fastembed embeddings at cosine > 0.99 before any Jacobian is trusted.

4.1 The problem an encoder poses

The causal J-lens is defined per token position toward next-token logits. An encoder emits a single pooled vector and no logits, so the per-position causal Jacobian is undefined here.

4.2 δ-broadcast averaged Jacobian

We perturb every source position by a shared displacement δ and differentiate the pooled output, collapsing the position×position Jacobian to one matrix per layer:

$$J_\ell \;=\; \frac{1}{T}\,\frac{\partial p}{\partial \delta} \;\in\; \mathbb{R}^{d\times d}.$$

J_ℓ is built by batched central finite differences ($\varepsilon = 0.05$, pure gemm), not per-dimension autograd. `lib.rs:layer_jacobian`. (Derivation: Appendix A.)

4.3 Embedding-space projection

Because p is L2-normalized, its radial direction carries no information, so we project:

$$J_{\text{emb}} \;=\; \tfrac{1}{\|p\|}\,(I - ee^\top)\,J_{\text{raw}}, \qquad e = p/\|p\|,$$

which is unit-tested to satisfy $e^\top(J_{\text{emb}}v) \approx 0$. `lib.rs:to_embedding_jacobian`.

4.4 Vocabulary (logit) lens

Standardized $J{\cdot}h$ times the tied WordPiece embeddings $W_U \to$ top tokens. Noisy here (the checkpoint ships no trained MLM head); reported but not load-bearing. `lib.rs:vocab_topk`.

4.5 Style lens

We read the Jacobian onto interpretable identity axes instead of the vocabulary:

$$s_a(h) \;=\; A_a \cdot \text{normalize}(J_{\text{emb}}\,h), \qquad A_a = \text{normalize}\,(\bar{c}_a^{+} - \bar{c}_a^{-}),$$

each axis a unit difference-of-class-means (Fisher) direction over the reference embeddings. Shipped axes, prose: `male` ↔ `female`, `educated=England` ↔ `rest`, `educated=US` ↔ `rest`, `raised=England` ↔ `rest`, `raised=US` ↔ `rest`; code: `Systems` ↔ `rest`, `Scripting` ↔ `rest`. `lib.rs:style_scores`, `axis`, `author_axes`.

4.6 J-space decomposition

Non-negative matching pursuit over unit rows of $W_U J_\ell$ yields a sparse concept set and the fraction of activation variance it captures. `lib.rs:jspace_nmp`.

4.7 Structural depth signatures

Per layer: stable rank $\|J\|_F^2/\sigma_1^2$, effective dimension (participation ratio $\|J\|_F^4/\|J^\top J\|_F^2$), verbalizability (excess kurtosis of the vocab-lens readout), layer×layer linear CKA, and, completing the paper's four-metric set, autocorrelation (lag-1 readout persistence vs. a position-shuffled null). `lib.rs:stable_rank`, `effective_dim`, `excess_kurtosis`, `linear_cka`, `readout_autocorrelation`.

5. Experiments and results

5.1 Identity is a computed intermediate

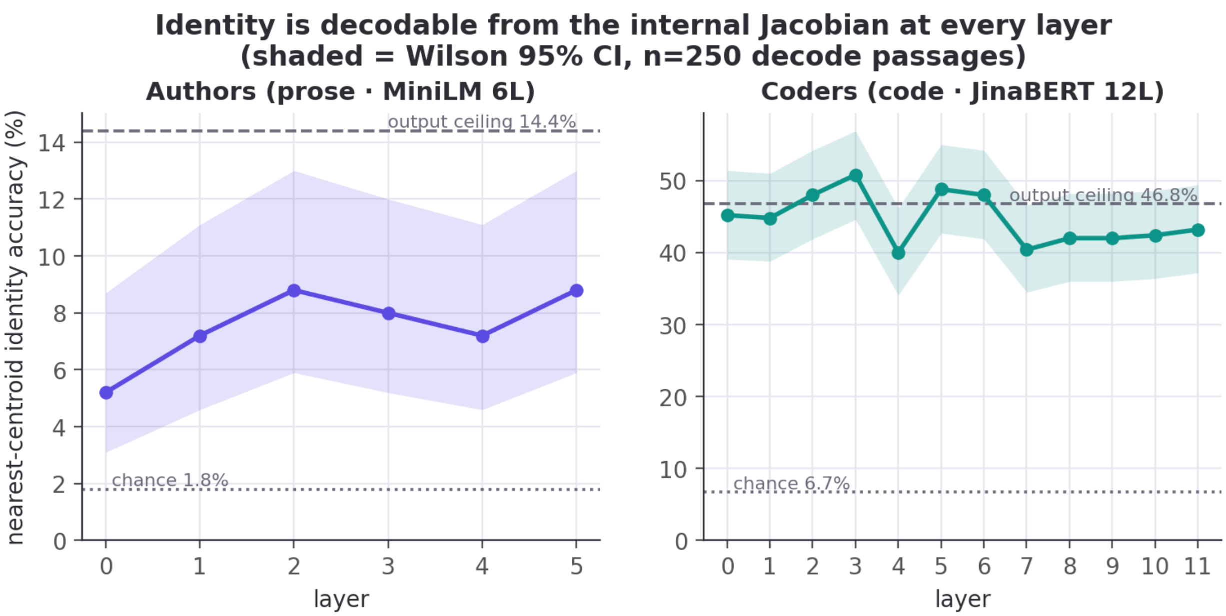

Figure 1: Identity accuracy decoded from the internal Jacobian at each layer, vs. chance (dotted) and the output-embedding ceiling (dashed).

Decoding author identity by leave-one-out nearest-centroid on the internal per-layer Jacobian readout beats chance at every layer. For prose, the per-layer rates $[5.2, 7.2, 8.8, 8.0, 7.2, 8.8]$% all clear the 1.8% chance (output ceiling 14.4%); even the weakest (5.2%, Wilson $[3.1, 8.7]$, n=250) excludes chance (binomial $p <$ 0.001). For code the rates run far higher, up to 50.8% against a 6.7% chance. The best internal code layer (50.8%, $[44.6, 56.9]$) matches the output-embedding ceiling (46.8%, $[40.7, 53.0]$): the intervals overlap by $\sim$9 points, so the internal readout is *as good as* the output, not better. Identity is computed inside the layers, not merely emitted. *(Wilson 95% CIs, decode sample n=250; per-layer values in ledger `identity_*`, bounds in `ci_identity_*`.)*

Does the Jacobian readout beat a plain probe? The natural baseline is to decode identity from the raw mean-pooled activation at each layer, with no Jacobian. On prose the two are indistinguishable: best layer 8.8% (J-lens) versus 8.4% (direct probe), and 7.5% versus 7.0% averaged over layers, both well inside the Wilson intervals above. The averaged Jacobian adds no decode accuracy over a direct probe, which bounds this contribution: the identity-at-every-layer result is robust to the readout, and the Jacobian's value is interpretability (the style lens onto named axes, §4.5) and causal steering (§5.4), not raw accuracy.

5.2 Identity ignition — does the space collapse to a single person?

We adapt the paper's ambiguous-input ignition to identity. For an author pair (A, B) we build a per-depth difference-of-means axis $\hat{u}_\ell = \widehat{c_{A,\ell} - c_{B,\ell}}$ from their *training* passages, then blend two *held-out* passages at the input embedding, $h_0(\alpha) = (1-\alpha)h_0^B + \alpha h_0^A$, sweep $\alpha \in [0, 1]$, and read the commitment $s_\ell(\alpha) = \hat{u}_\ell(\bar{p}_\ell(\alpha) - m_\ell)$ at every depth (15 pairs). A graded layer ramps linearly in α; an *ignited* layer snaps.

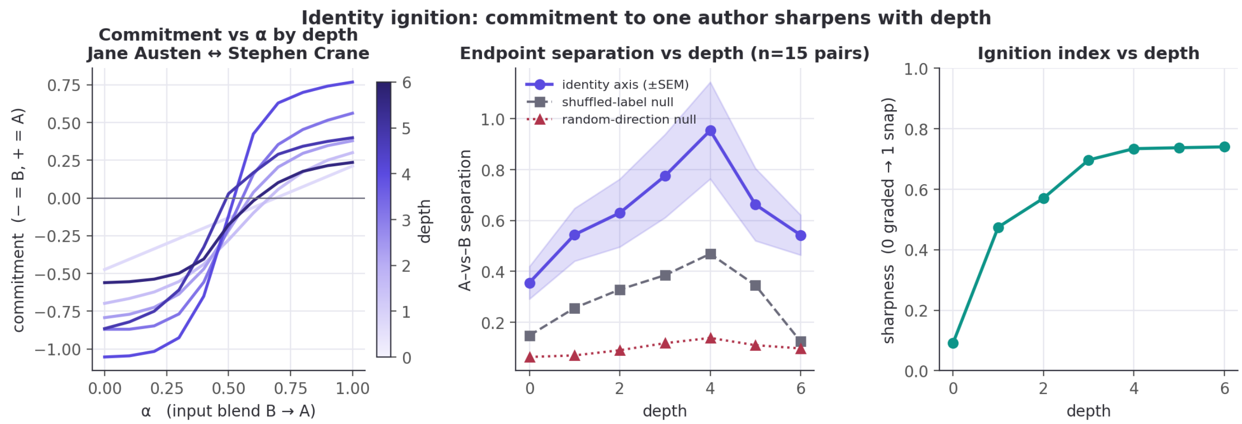

Figure 2: Identity ignition (MiniLM). Left: commitment vs. α by depth for one pair (input-linear at depth 0, stepped by mid-depth). Middle: A-vs-B separation rises to a mid-network peak, far above the random null (~7×) and clearly above the shuffled null. Right: the ignition index (transition sharpness) rises with depth.

The representation commits to one author, increasingly with depth. Endpoint separation along the identity axis rises from 0.36 at the input to a mid-network peak of 0.95 at depth 4 (of 6; ±0.19 SEM over 15 author pairs), then eases to 0.54 at the output. It dominates both controls at every depth: the random-direction null sits at 0.06–0.14 (a $\sim 7\times$ margin at the peak) and the shuffled-label null at 0.12–0.47. So the effect is identity-specific, not a generic consequence of blending inputs. The ignition index (transition sharpness, $0 =$ graded, $1 =$ all-or-none) climbs from 0.09 at the input (where the readout is, correctly, linear in the blended input) to 0.73–0.74 by mid-depth and holds. In short: *shallow layers hold a graded mixture; by the middle of the network the representation has snapped to a single author.* The mid-network peak echoes the low-rank "workspace-like" bottleneck we see structurally (§5.3).

The same holds for code, more strongly. On the 12-layer code encoder the ignition index climbs from 0.09 at the input to a peak of 0.82 (depth 11 of 12). Through the early-mid layers the identity axis separates developers far above the random-direction null: a $\sim 13\times$ margin at depth 4 (1.21 vs 0.09; raw separation peaks slightly earlier, 1.56 ± 0.22 at depth 3). It is a sharper version of the same effect, consistent with code identity being more linearly accessible overall (§5.1). Separation spikes higher still at the final layer (2.33), but that depth is noisy, its wide ±0.53 SEM and a jumping null are why we feature the stable early-mid layers. Both modalities show the representation committing to one individual with depth.

Two caveats bound this reading. It rests on a 6-layer encoder and 15 pairs, and although the sharpening is measured *along an axis the separation control proves is identity-specific*, we cannot fully exclude that some of the depth-wise sharpening reflects generic late-layer nonlinearity. We report the raw depth series.

5.3 Structural signatures across depth

Effective dimension rises with depth in both models (prose $73 \to 205$; code $80 \to 185$). The stable rank tells the sharper story: in the 6-layer prose model it is lowest at the very first layer (22.1, rising to 106.9), so the network compresses to a low-rank readout immediately, whereas the 12-layer code model has a mid-network low-rank bottleneck (minimum 20.3 at layer 5 of 12, with a matching dip in effective dimension). The mid-network "workspace" geometry the paper reports in deep models appears here only once there is depth to spare. The fourth signature, autocorrelation (persistence of the readout across positions), completes the

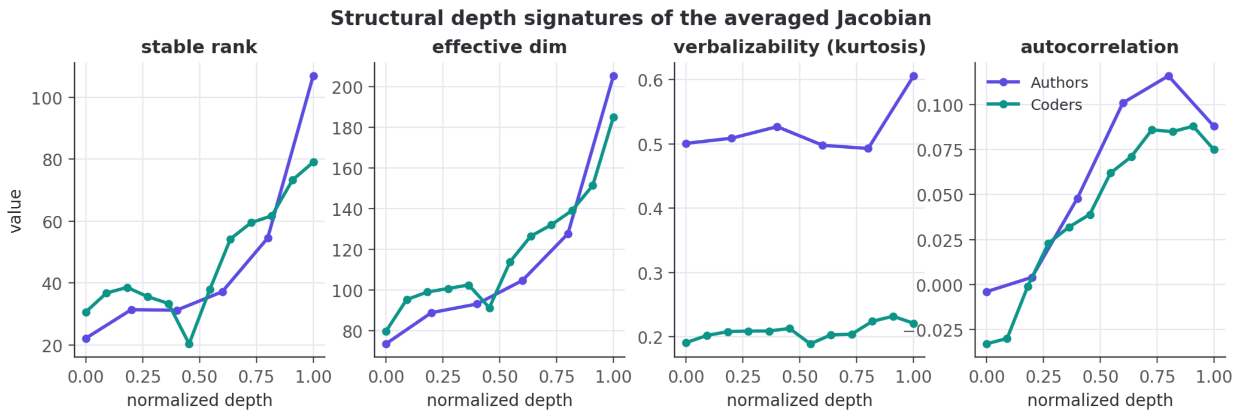

Figure 3: Structural depth signatures of the averaged Jacobian (normalized depth).

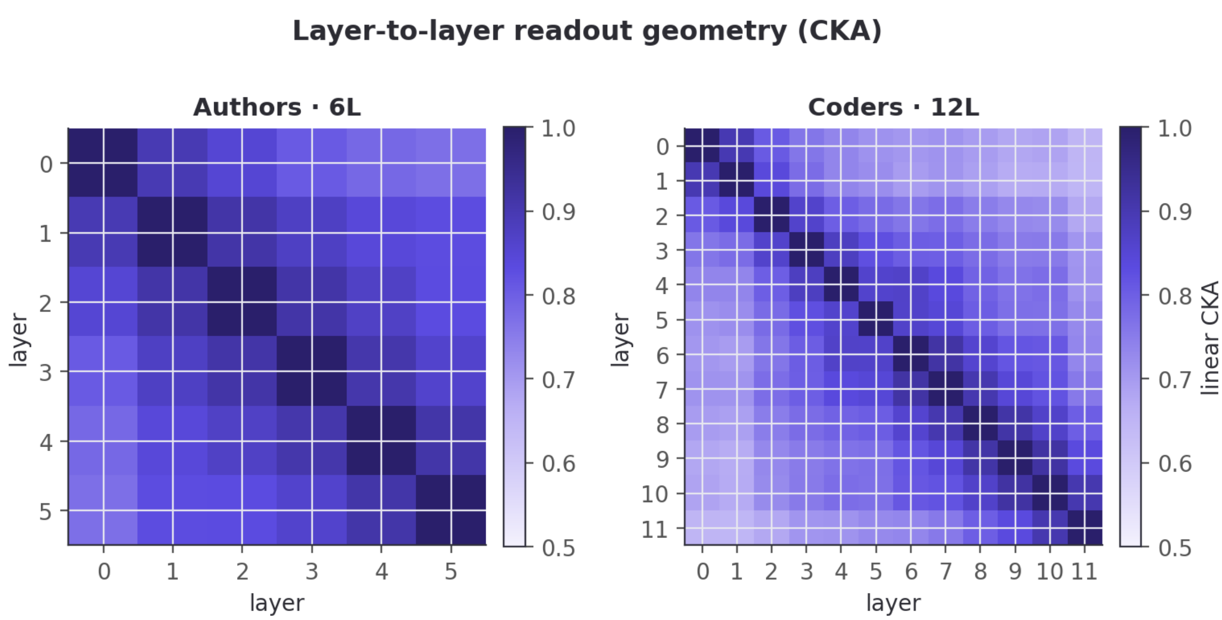

Figure 4: Layer-to-layer readout geometry (linear CKA).

picture: near zero or negative at the shallowest layers and rising through the middle (prose peaks at 0.12 around layer 4; code climbs to ≈ 0.09 by layers 8–9), the workspace-persistence signature of (Anthropic 2026) surfacing even on these shallow encoders. We do not see the paper's clean sensory→workspace→motor tripartition (6–12 layers is too shallow), and we report the raw depth series rather than forcing that reading.

5.4 Fingerprint presence and causal steering

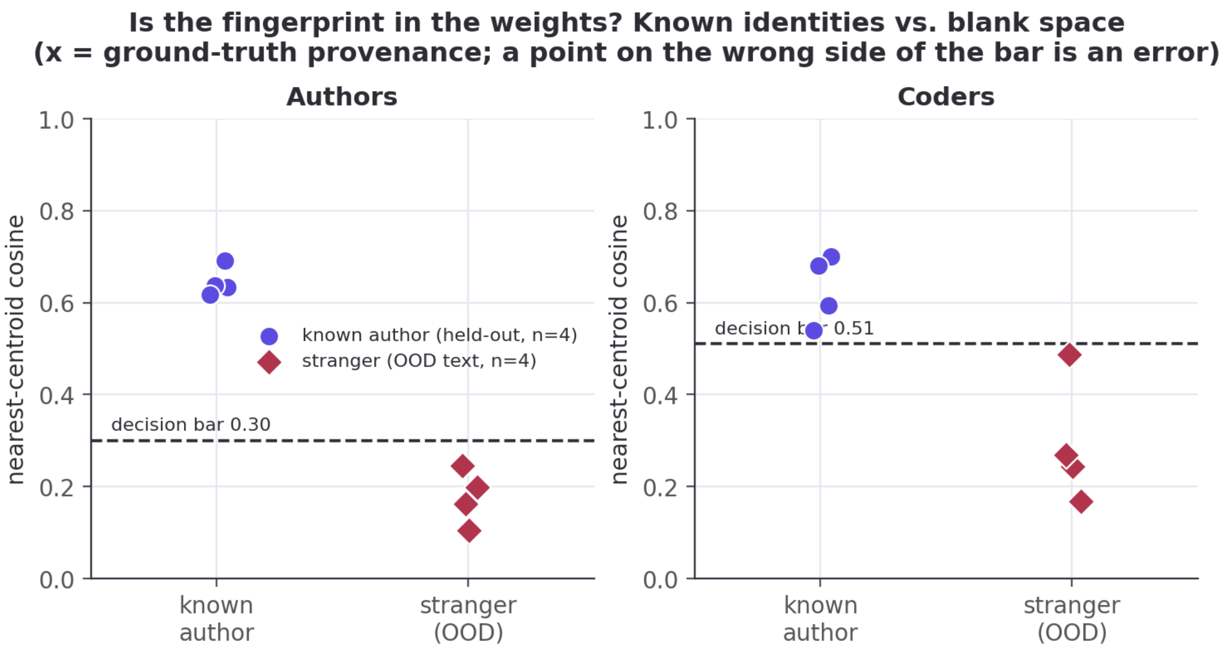

Figure 5: Is a person's fingerprint in the weights? Known identities vs. out-of-distribution "blank space".

A calibrated nearest-centroid detector separates known identities from "blank space". Prose (bar 0.30): known probes 0.62–0.69, out-of-distribution text (code, chat, biology, legalese) 0.10–0.25. Code is tighter (bar 0.51): known 0.54–0.70, OOD 0.17–0.49; "modern chat" sits just under the bar.

The identity direction is a causal lever, not just a readable one. Forming the residual steering direction $\hat{\delta} = \widehat{J_{\mathrm{emb}}^{\top} A}$ and injecting $\alpha\hat{\delta}$ at the mid layer drives the output's loading on axis A through a large, sign-controllable swing that runs monotonically from negative (at $\alpha = -6$) through zero (unperturbed) to positive (at $\alpha = +6$), saturating (and for gender slightly reversing) at the extreme α. The effect has the same sign across all five prose identity axes (gender, education, upbringing), though not the same magnitude: the gender axis swings least (0.80), the other four up to 1.11. A matched-norm random direction, injected identically, leaves the loading essentially flat (total drift ≤ 0.16 over the same sweep). So the layer holds the identity direction as something the rest of the network *acts on*, not merely correlates with. One caveat: the steer is built from axis A and read back on the *same* axis A, so the readout is not fully independent of the intervention; the matched-norm random control bounds how much of the swing that shared axis could manufacture (drift ≤ 0.16 vs. swings up to 1.11), and §5.6 gives the leakage-free, independent-readout analogue on a decoder.

5.5 Decoder track — does the structure hold on a generative model?

Everything above is on *encoders*. If the depth-wise workspace geometry is a real property of the J-lens apparatus and not an artifact of masked-mean pooling, then reading a generative decoder with the *same* averaged-Jacobian machinery should reproduce the paper's Figure-28 depth signatures. Hypothesis: on an open decoder we will see (i) the four Jacobian signatures vary with depth, and (ii) a decoder-only signature, next-token logit-lens accuracy, rise sharply in the late layers, marking the "motor" regime where representations turn toward the output. We do not expect the clean sensory→workspace→motor tripartition: GPT-2 is 12 layers, far short of the paper's ~100, so we report the raw series and let it say what it says.

Method. GPT-2 (Radford et al. 2019) (`openai-community/gpt2`; 12 layers, d=768), the δ-broadcast averaged Jacobian made causal (perturb every position, read the *last* position, the next-token driver) over prose prompts, with the four signatures computed by the *same* `lib.rs` functions as the encoders and verbalizability read through GPT-2's real tied unembedding (no approximation). This is the paper's mechanistic apparatus on an open model that actually generates.

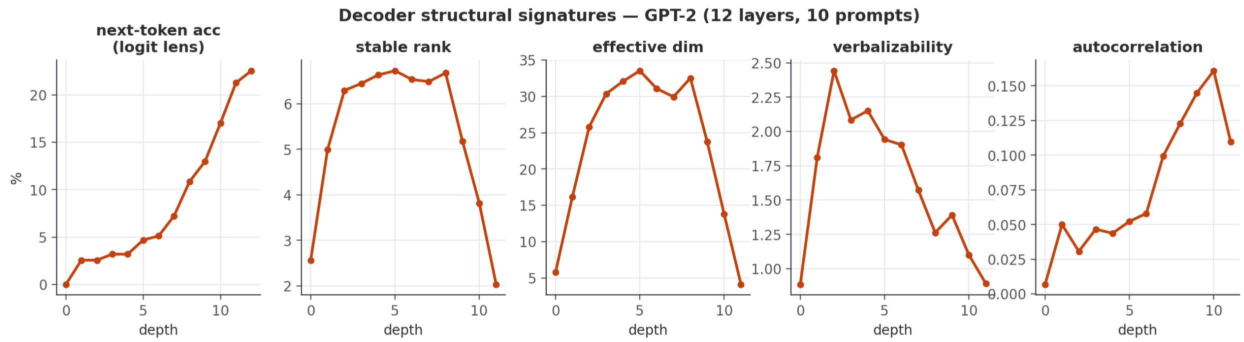

Figure 6: Decoder structural signatures on GPT-2 (Figure-28 series + next-token accuracy).

Result: both predictions hold, and more cleanly than on the encoders. Next-token accuracy is near zero through the first six layers and then climbs steadily to 22.6% at the output, hypothesis (ii): the late layers are the motor regime. The Jacobian's stable rank and effective dimension trace a pronounced inverted-U: low at the input (2.6 / 5.8), high through the middle (6.7 / 33.5 at layer 5), collapsing to near rank-one at the output (2.0 / 4.1), a sensory→workspace→motor signature that is *sharper on the 12-layer decoder than on the 6-layer encoders*, just as the "needs depth to spare" reading predicts (hypothesis (i)). Autocorrelation rises with depth to a peak of 0.16 (layer 10) before easing to 0.11 at the output, echoing the encoder workspace-persistence.

Two caveats bound this. Final-layer next-token accuracy (22.6%) is low (archaic literary prose, a 48-token context, 10 prompts), so read the accuracy *shape*, not its level. And our motor-end *collapse* is the opposite of the paper's motor behavior (there $J_\ell \to$ identity, full rank): the difference is deliberate and methodological, since our decoder Jacobian reads the last position (the next-token driver), which naturally becomes low-rank as the network commits to a single output, rather than the full residual stream the paper differentiates. The workspace geometry transfers; the motor *readout* is ours, and we flag it as such.

5.6 Behavioral steering — is the lever causal on a decoder?

The paper's boldest claims are behavioral: swap a J-lens vector for a reasoning intermediate and the answer changes. We wanted to test the strongest version (take a prompt the model answers wrongly and steer it right), but on a 124M-parameter GPT-2 that barely reasons, that test is neither clean (steering toward the answer token is just injecting the answer) nor likely to succeed. So we test the mechanistic prerequisite instead, and report its ceiling. Hypothesis: injecting a *concept-context* direction (built leakage-free as the difference of mid-layer mean residuals between concept-primed and neutral prompts, not the target's unembedding) raises concept-related tokens in a held-out neutral prompt, more than a matched-norm random direction.

Result: the lever is causal and bidirectional, but modest. Injecting $+\alpha\hat{\delta}$ at the mid layer raises the mean log-prob of held-out concept tokens by +0.10/+0.22/+0.44 (money / music / war), and $-\alpha\hat{\delta}$ lowers it below baseline in every case; the mean effect is +0.254 for the real direction versus −0.085 for the matched-norm random control. So a concept direction extracted purely from context is a sign-controllable causal handle on the decoder's output distribution, the prerequisite the paper's behavioral experiments rely on.

This is the prerequisite, not the headline. The effect is small: it shifts the *distribution* (raising concept-token log-probability by +0.25 on average; mean Δ, with a matched-norm random control at ≈ 0) but does not flip the model's actual output, and it rests on only 3 concepts $\times$ 6 prompts. It is not the paper's "can't→can" reasoning result, and we do not claim it is. Redirecting a *reasoning intermediate* to change a final answer needs a model that can reason; GPT-2 shows the handle exists and is causal, and marks where a capable open decoder (e.g. Qwen2.5) is required to go further. We regard that as the next step, not a result we have.

5.7 Measured constructs, a negative control, and why we make no "IQ" claim

Every attribute so far (gender, region, systems-vs-scripting) is a *label we assigned*. The sharpest test of whether these embeddings carry real psychological signal is to hand them a population whose attributes were measured by someone else, with a validated instrument, and ask whether the signal survives. Hypothesis: if prose style encodes personality at all, an embedding built with no knowledge of psychology should recover self-reported Big Five traits above chance. Method: the Pennebaker & King stream-of-consciousness corpus (Pennebaker and King 1999), 2,467 essays, each labelled with the writer's Big Five traits from a self-assessment questionnaire (ground truth, not our guess), embedded with the *same* MiniLM used throughout. For each trait we run leave-one-out nearest-centroid classification (high vs. low group) against the majority-class baseline, with a shuffled-label null.

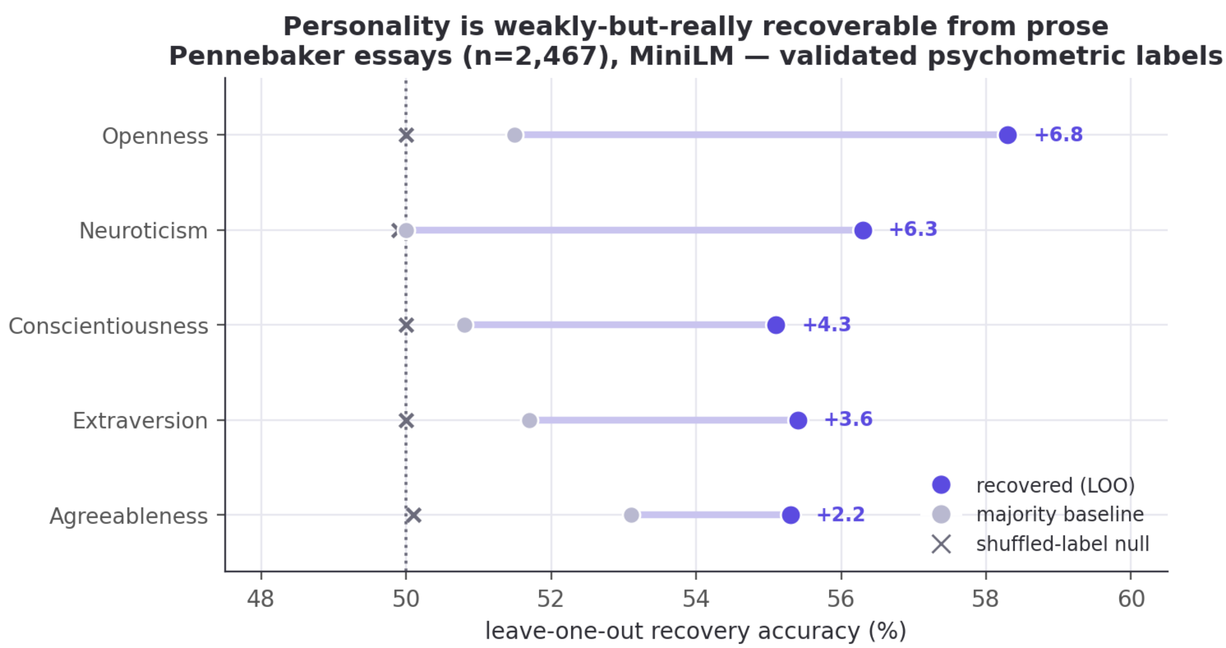

Figure 7: Big Five recovery from prose: leave-one-out accuracy vs. majority baseline, with the shuffled-label null flat at chance.

Result: every trait beats a shuffled-label null, and the null is flat. Openness is strongest at 58.3% (majority 51.5%; +6.8 points), then Neuroticism 56.3% (+6.3), Conscientiousness 55.1% (+4.3), Extraversion 55.4% (+3.6), Agreeableness 55.3% (+2.2), a mean lift of +4.6 points over the majority baseline. A 1000-permutation test, which shuffles the trait labels and re-runs the whole classifier, puts all five traits at $p < 0.001$ against that null (which sits at chance, 49.9–50.1%): the classifier is extracting real trait signal, not fitting noise. The permutation test is against label-shuffling, not the majority baseline, so we also check each trait's Wilson 95% interval against its majority class: Openness clears it comfortably (58.3%, $[56.3, 60.2]$ vs. 51.5%), but Agreeableness only marginally (55.3%, $[53.3, 57.2]$ vs. 53.1%), its lower bound barely exceeding the baseline. The lift over majority is weak for the lowest traits, as expected. This is the *weak-but-real* ceiling the personality-from-text literature reports (Mairesse et al. 2007), and Openness-leads-the-pack is a standard finding at social-media scale (Schwartz et al. 2013). It is a positive result on a measured population: author2vec, trained for nothing of the kind, carries a faint but real trace of who the writer is.

A negative control: what the method does *not* recover. A positive result could still, in principle, be spurious structure. The decisive test is to pair it with an attribute that *should* be unrecoverable and confirm the method stays silent. The Blog Authorship Corpus (Schler et al. 2006) labels every blogger with gender, age, and astrological sign: the first two have linguistic correlates, the third has none. We pool posts to the author level (138 authors with a known sign; each author is the mean of their post embeddings) and run the *identical* leave-one-author-out nearest-centroid classifier on all three, each against a shuffled-label null.

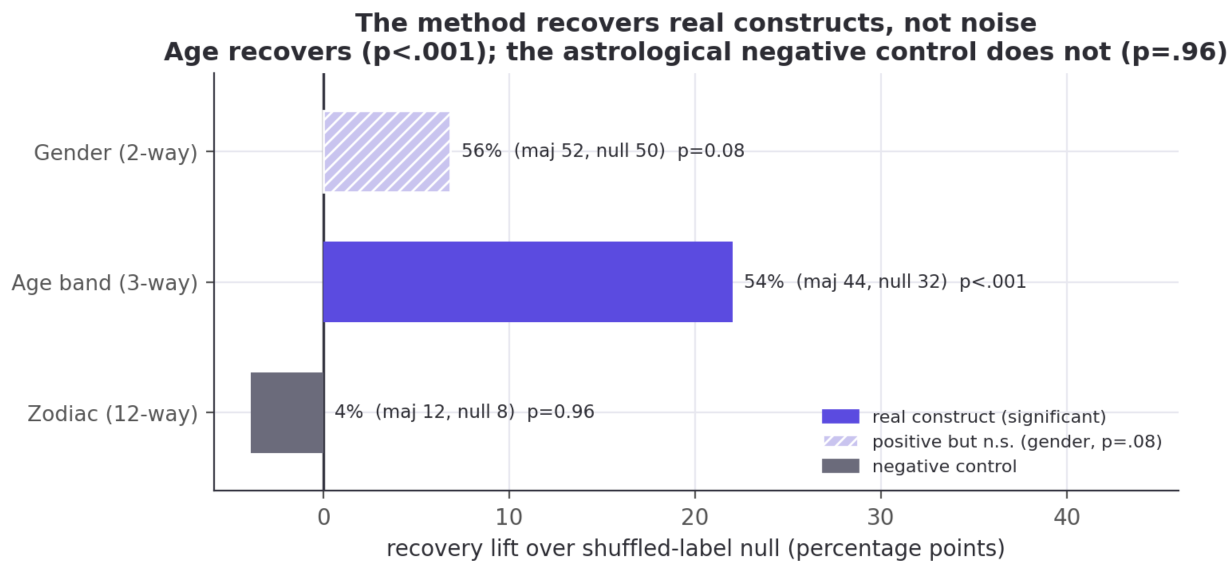

Figure 8: Construct validity on one corpus (author-level, 1000-permutation test): writer age recovers ($p < 0.001$) and the astrological-sign negative control does not ($p = 0.96$); gender is directionally positive but not significant at this sample ($p = 0.08$).

Age band recovers strongly and significantly: 53.6% (majority 43.5%) against a 31.6% shuffled-label null, $p < 0.001$ over 1000 permutations, while astrological sign shows *nothing*: 4.3%, *below* both its majority baseline (11.6%) and its shuffled null (8.2%), $p = 0.96$ (the real labels do worse than 96% of random relabelings). Gender lands in between, directionally positive but not significant at this modest sample (56.5% against a 49.7% null, $p = 0.08$), consistent with a weak signal that 138 author-level points are underpowered to confirm. The contrast is unambiguous: the *exact* pipeline that recovers a construct with a linguistic basis (age) finds nothing in the astrological control. This is the boundary a "measured population" is *for*: evidence that the recovered signal is real where a real construct exists, and absent where none does.

Those two results together are why we make no "IQ" claim. A validated psychometric construct tops out a few points over chance; a loaded, poorly-operationalized one like "intelligence" would fare no better and would invite far worse misreading. A companion check makes the point concretely: if the recovered *education* style axis were a proxy for lexical sophistication, an author's projection onto it should correlate with concrete lexical metrics, yet over all 55 authors it yields only weak positive correlations, $r = +0.10$ (mean word length), +0.11 (type-token ratio), +0.20 (long-word fraction). Even a labelled, human-interpretable identity axis only faintly tracks a concrete lexical proxy. Whether one can steer a *capability* score (a vocabulary test administered to a decoder) is a question for a capable model with careful construct validation. That is future work, not a result we have, and we make no intelligence claim.

5.8 The authorship study (context) & style trajectories

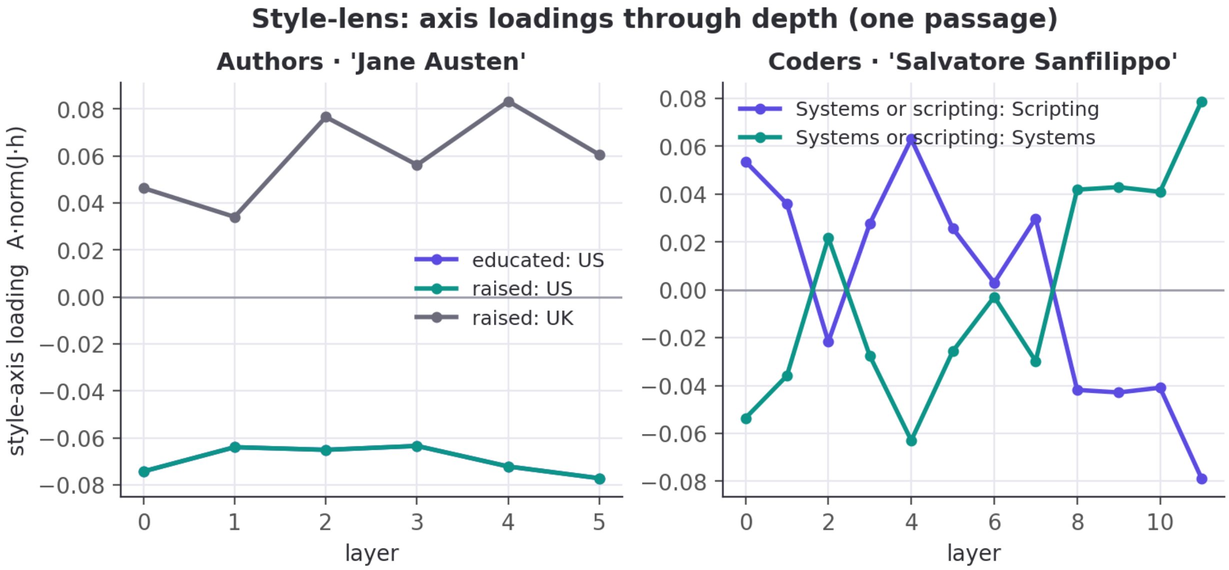

Figure 9: Style-lens axis loadings through depth for one passage.

Downstream, the same embeddings support an authorship study: prose recognition climbs from a 1.8% blind baseline to 58.3% once the model has read the author's book; a new-passage reveal is 130/220 correct when the author is known and 0/220 when fully hidden (code, 15 developers: 6.7% → 71.1%; reveal 45/60). Trait recovery is strong for some traits and absent for others: prose gender 85.5% (majority 52.7%) but most geographic traits at or below their majority baselines; code systems-vs-scripting 80.0% (majority 53.3%) but commit-time and weekend near or below chance.

5.9 Does the model know some coders better than others? And is style homogenizing?

The 15-developer code roster lets us ask two questions the prose side cannot. (That developers are identifiable from code style at all is established: code stylometry de-anonymizes programmers (Caliskan-Islam et al. 2015); our question is instead *where in the network*, and *how much*, that identity is legible.)

Some developers are far more recognizable than others. Same-file recognition accuracy (Wilson 95% CIs, n=126 files/developer) spans a wide range, from Jeremy Ashkenas at 89.7% [83.1, 93.9] (a highly idiosyncratic CoffeeScript/Backbone style) down to Armin Ronacher at 48.4% [39.9, 57.1], all far above the 6.7% chance floor. The top and bottom are separated: Ashkenas's and Ronacher's intervals are disjoint, so the ~41-point extreme spread is real, not sampling noise. The *fine* ranking is not: the two developers we added to probe the tails, Andrew Kelley (Zig, 72.2% [63.8, 79.3]) and Jarred Sumner (Bun, 54.0% [45.3, 62.4]), have intervals that overlap much of the middle of the roster, so we read only the *coarse* structure (top vs. bottom third) as reliable, not any individual's exact rank. We report this as *measured recognizability* and resist over-reading it: a lower score reflects how stylistically distinctive a developer's *attributed, name-scrubbed* code is in this corpus, confounded by codebase heterogeneity (Bun mixes Zig, C++, and generated code across many hands), not a verdict on the person.

No sign of AI-era style homogenization. If a shared external influence (AI assistants) were melting coders into one style, cross-coder similarity should *rise* in the AI era. Time gives a clean hold-out: pre-2021 commits are provably AI-free (Copilot shipped mid-2021, ChatGPT late 2022). Comparing mean cross-coder style similarity pre-2021 vs. 2023-onward, over the 9 developers with enough history in both eras, the change is +0.005 (0.568 → 0.574), indistinguishable from a 500-sample within-coder permutation null (null mean

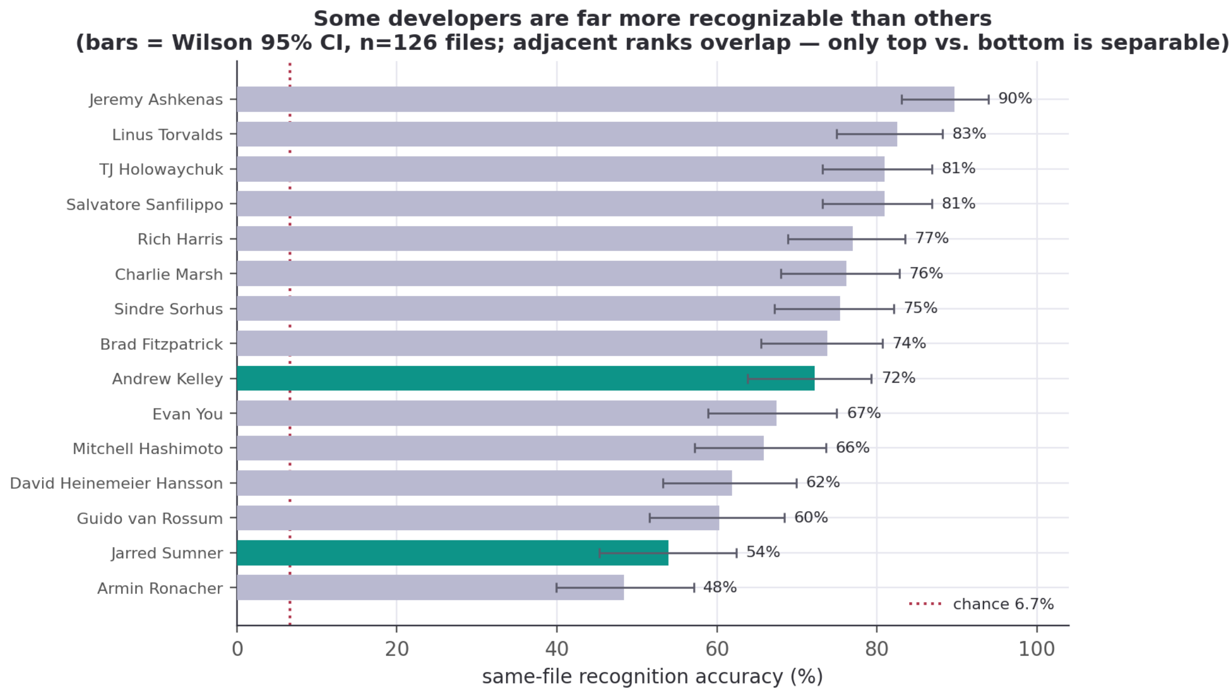

Figure 10: Per-coder same-file recognizability across the 15 developers (Kelley and Sumner highlighted); chance 6.7%.

-0.033, two-sided $p = 0.96$), while each developer stays recognizably themselves across the boundary (within-coder self-consistency 0.77). We find no evidence of homogenization. Whatever AI assistants are doing to code, they are not, on this roster, collapsing individual style. We make no claim about which individuals do or don't use AI: the population signal is null, and per-developer attribution would be both unsupported (confounded, no ground truth) and inappropriate.

5.10 Can we fingerprint the AI models? — the task dominates, a faint model signal survives

If humans have fingerprints, do the *models* people code with? Here we finally have ground truth: we generate the code, so we know which model wrote it. We prompted four latest frontier models (`claude-opus-4.8`, `gpt-5.6-terra`, `gemini-3.5-flash`, `deepseek-chat`) across 40 coding tasks (24 canonical + 16 open-ended) and embedded the output in the same JinaBERT space as the humans (149 of the 160 model×task cells produced non-empty code; the rest were empty generations).

A control rules this out. Same-task, different-model code is alike (cosine 0.71), which *looks* like "the frontier models have converged to one style." We do not make that claim, because a control refutes it: same-model / different-task similarity is only 0.20, so it is the task, not the model, that drives the embedding. On a canonical "implement an LRU cache" every model writes the textbook answer, so the high cross-model number mostly measures *task* overlap, not model identity. Controlling for the task, leave-one-task-out nearest-model-centroid still recovers which model wrote unseen code at 42% against a 25% chance baseline, a faint fingerprint, not convergence.

The fingerprint is real but faint. That 42% recovery ($[34.6, 50.3]$ Wilson, n=149) is significantly above a within-task label-shuffling null (null mean 24%, $p < 0.001$ over 1000 permutations), so the models *do* carry a detectable style, but it is largely masked by what the code *does*. The summary: authorship is strongly recoverable for humans (48–90%) and only weakly for models, and for models it is the task, far more than the author, that shapes the code. No convergence claim; no individual-usage claim.

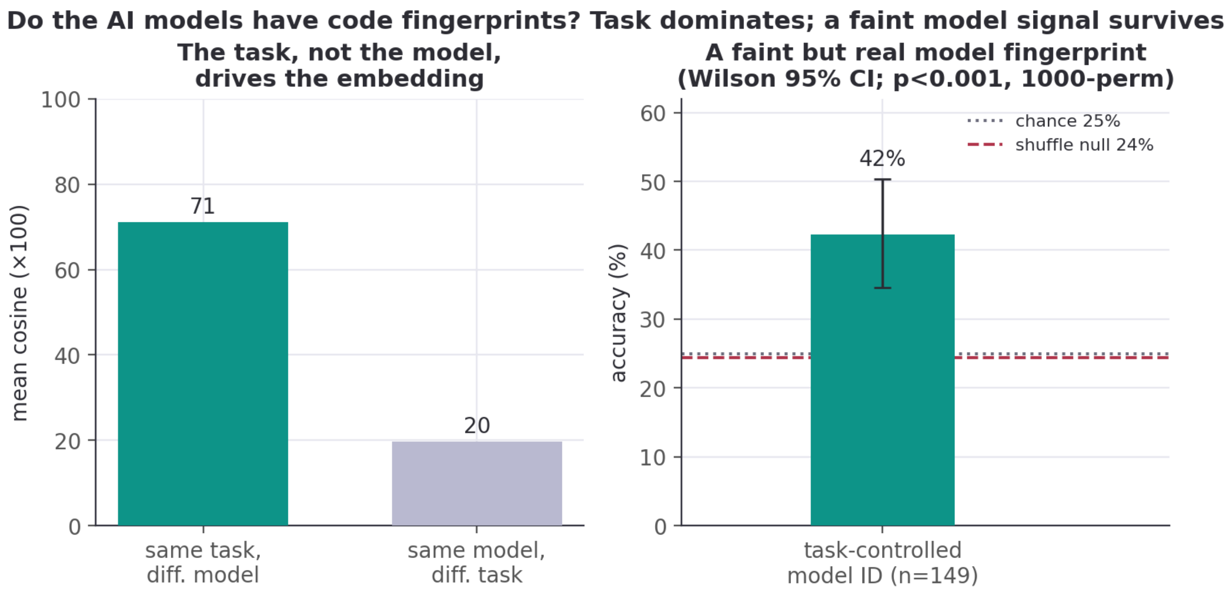

Figure 11: Task vs. model: on the same task, different models write nearly the same code (0.71); the same model across different tasks is far less alike (0.20). The task, not the model, dominates the embedding; controlling for it, model identity is recoverable at 42% vs 25% chance.

5.11 Ablations

Rather than a hyperparameter sweep, the study carries several *built-in* ablations. Exposure / hold-out: the familiarity ladder (§5.8) is a leave-one-{book,series,author}-out ablation of how much of a subject the model has seen, shipped in `results`. Depth: the 6-layer prose vs. 12-layer code encoders act as a depth ablation, with the mid-network workspace bottleneck (§5.3) and the ignition peak (§5.2) pronounced only in the deeper model. Model class: the GPT-2 decoder (§5.5) ablates encoder → generative on the *same* apparatus, and the workspace geometry survives. Controls-as-ablations: every causal claim ablates its direction against a matched-norm random and/or shuffled-label null (§5.2, §5.4, §5.6). What we did *not* sweep, finite-difference ε and Jacobian context length, we flag as future work; the signatures reproduce bit-for-bit at the documented settings, so the qualitative curves are stable.

6. Discussion

What the identity-workspace analogy supports. Read with the same averaged-Jacobian apparatus as (Anthropic 2026), an embedding encoder does exhibit workspace-like *geometry* around identity: the fingerprint is computed internally at every layer (§5.1), it becomes increasingly linearly separable and commits to a single individual with depth (§5.2), the readout compresses through a low-rank mid-network bottleneck (§5.3), and the identity direction is a causal lever (§5.4). On a real decoder the same signatures sharpen into a clean sensory→workspace→motor profile (§5.5). The *mechanistic* half of the paper's picture transfers, and it transfers to a question the paper never asked: *whose style is this?*

What it does not support. None of this is evidence of reportability, reasoning, or anything cognitive. Our decoder steering moves a distribution, not an answer (§5.6); our interpretable axes barely track even lexical sophistication (§5.7). The "workspace" here is a claim about representational geometry and linear accessibility, not about a model *knowing* or *reporting* who you are. The defensible reading is the narrow one: personal writing style is a low-dimensional, causally-active, depth-localized direction in these models' representations, which is both less than the slogan "the AI knows you" and more precise than it, because it is checkable on open models.

Why an encoder was the right testbed. Stripping away generation removes the decoder story's main confound (autoregressive leakage) and lets the mechanistic claims stand or fall on geometry alone. That the same geometry then reappears, more cleanly, on a 12-layer decoder (§5.5) is the strongest cross-check we can offer at this scale.

7. Limitations and threats to validity

Replication vs. novel. The J-lens, J-space decomposition, and the four structural signatures are the paper's (Anthropic 2026); our contribution is the encoder adaptation, the style-lens readout, the identity target, and reproducibility. We are careful not to claim the apparatus.

Construct scope. We measure *identity commitment* and (§5.7) an *expertise/lexical register*, not IQ, not consciousness. We borrow the *ignition* experimental design, not the conclusion.

The ignition result needs its caveat stated plainly. The ignition index rising with depth (§5.2) is measured *along an axis the separation control proves is identity-specific* (real separation runs $\sim$7–13$\times$ above a random-direction null). But we cannot fully exclude that *some* of the depth-wise sharpening is generic late-layer nonlinearity: any readout of a linearly-blended input can become more nonlinear with depth. What the controls establish is that the *axis* carries identity; the raw sharpening curve should be read as suggestive, not decisive. Our shuffled-label null is also imperfect (its two groups still contain real passages, so it sits above zero); the random-direction null is the cleaner floor.

Statistical power. Ignition uses 15 author/coder pairs and a 7- or 11-point α grid; structural signatures average over 32–96 passages. These are small, and the ignition *index* at each depth averages only over the pairs whose endpoints separate along the axis (5–14 of 15, fewest at the shallowest layers). We report standard deviations and the raw per-depth series rather than smoothed summaries.

Reproducibility caveats. Encoder structural signatures reproduce bit-for-bit on MiniLM across runs; the deeper JinaBERT numbers differ from an earlier site build generated with different (undocumented) settings; we report the values from a run with documented settings (`JLENS_JAC_LEN=64`) and have not re-verified GPU-reduction determinism on the 12-layer model. The single-averaged, single-direction lens is lossy; the vocab lens is noisy on encoders (no trained MLM head); finite-difference ε introduces $O(\varepsilon^2)$ error.

Scope. 55 authors / 15 coders on laptop-scale models demonstrate the *mechanism*; that a frontier model trained on someone's millions of words holds a sharp, personal fingerprint of *that individual* is a well-motivated extrapolation, not something these data settle. The behavioral half of the paper (§5.6–5.7) is the hardest to carry over and is where a small open decoder's limited capability bites; we state results there as directional, with negative results reported as such.

Content vs. style (prose). Our public-domain authors write about different subjects, so some prose "identity" signal is inevitably topic, not style (Wegmann et al. 2022). Unlike the code side, where we control the task explicitly (§5.10), we do not fully disentangle the two on the prose side; prose identity should be read as *style-or-topic* fingerprinting, not pure style.

Ethics and broader impact

This work shows that authorship identity and correlates of demographic and psychological attributes (gender, region, age, Big Five personality) are linearly recoverable from prose and code embeddings, and that these directions are causally steerable. The apparatus is dual-use: the same machinery can support de-anonymization of pseudonymous authors, including developers from name-scrubbed code (Caliskan-Islam et al. 2015; Narayanan et al. 2012), and non-consensual inference of sensitive attributes, enabling surveillance or profiling (Hovy and Spruit 2016). We take three positions. Consent and scope: our prose subjects are public-domain authors and our developer corpus is public, git-attributed, and name-scrubbed; we make no per-individual attribution claim and explicitly decline the AI-usage and "intelligence" inferences the method could invite (§5.7, §5.9). Construct restraint: the astrological-sign negative control ($p = 0.96$) is included to show that a recovered signal must be validated against a real construct before use, and to caution against reading weak correlational demographic signals as ground truth. Mitigation: the demographic directions we find can also be *removed* (nullspace projection / INLP (Ravfogel et al. 2020)), and we regard attribute erasure and stylistic obfuscation as the appropriate defensive counterpart to this analysis. We release only open models and public corpora, and no tool aimed at identifying specific private individuals.

8. Reproducibility

Models. `sentence-transformers/all-MiniLM-L6-v2` (prose; 6 layers, d=384) and `jinaai/jina-embeddings-v2-base-code` (code; 12 layers, d=768), loaded natively via candle and gated (`bin/spike`) to reproduce the shipped fastembed embeddings at cosine > 0.99.

Pipeline.

```
cargo run -p corpus --release                 # prose corpus -> person2vec-minilm.{json,bin}
cargo run -p corpus --bin coders --release    # code corpus  -> person2vec-coders.{json,bin}
cargo run -p jlens  --bin spike     --release              # faithfulness gate
cargo run -p jlens  --bin jlens     --release -- <ds> <N>  # -> person2vec-jlens-<ds>.json
cargo run -p jlens  --bin steer     --release -- <ds>      # -> person2vec-identity-<ds>.json
cargo run -p jlens  --bin fingerprint --release -- <ds>    # -> person2vec-fingerprint-<ds>.json
cargo run -p jlens  --bin ignition    --release -- <ds>    # -> person2vec-ignition-<ds>.json
 ↪  (sec 5.2)
cargo run -p jlens  --bin steer_bundle --release -- <ds>   # -> person2vec-steer-<ds>.json
 ↪  (sec 5.4)
cargo run -p jlens  --bin decoder_structural --release -- 10  # -> decoder-structural-gpt2.json
 ↪  (sec 5.5)
cargo run -p jlens  --bin decoder_steer --release          # -> decoder-steer-gpt2.json
 ↪  (sec 5.6)
python3 jlens/paper/expertise_probe.py        # -> person2vec-expertise-minilm.json
 ↪  (sec 5.7)
./target/release/embed_texts minilm essays_in.json essays_out.json   # embed Pennebaker essays
 ↪  (sec 5.7)
python3 jlens/paper/recover_bigfive.py        # -> person2vec-bigfive.json  (Big Five recovery)
 ↪  (sec 5.7)
python3 jlens/paper/recover_blog_demographics.py  # -> person2vec-blog-demographics.json  (neg.
 ↪  control, sec 5.7)
python3 jlens/paper/build_ledger.py           # -> jlens/paper/ledger.json  (every cited number)
python3 jlens/figures/make_figures.py         # -> jlens/figures/*.png
```

`<ds>` $\in$ {`minilm`, `coders`}. Env vars: `JLENS_DEVICE` (`metal`|`cpu`), `JLENS_JAC_LEN` (Jacobian context; default 64 for the encoder binaries, 48 for `decoder_structural`), `JLENS_CHUNK` (finite-difference batch; smaller = less GPU memory, identical numbers). Determinism: the structural signatures reproduce bit-for-bit across runs (verified); the ignition null uses a fixed splitmix64 seed. The `/jlens` viewer does no linear algebra; all quantities are precomputed offline and shipped as static JSON.

Auditability. Every quantitative claim resolves through the audit ledger (Appendix D, `jlens/paper/build_ledger.py`) to a source bundle and JSON path; every method claim resolves through `jlens/paper/method_map.md` to a `file:line`.

Appendix A — δ-broadcast derivation

A masked-mean-pooled encoder outputs $p = \frac{1}{T}\sum_t h_{L,t}$, the mean over T positions of the final layer. The full sensitivity of p to the layer-ℓ residuals is a rank-4 object $\partial p_i / \partial h_{\ell,t,j}$ of size $d \times (T \times d)$, intractable to form and to average across variable-length prompts. We collapse it with a shared perturbation: perturb *every* source position by the same $\delta \in \mathbb{R}^d$, $h_{\ell,t} \mapsto h_{\ell,t} + \delta$, and differentiate the pooled output,

$$J_\ell \;=\; \frac{\partial p}{\partial \delta}\Big|_{\delta=0} \;=\; \frac{1}{T}\sum_t \frac{\partial p}{\partial h_{\ell,t}} \;\in\; \mathbb{R}^{d\times d},$$

one $d \times d$ matrix per layer, independent of T. Each column c is estimated by central finite differences, $J_{\cdot c} \approx \frac{p(+\varepsilon e_c)-p(-\varepsilon e_c)}{2\varepsilon}$ (with $\pm\varepsilon e_c$ broadcast to all positions), whose error is $O(\varepsilon^2)$ by Taylor expansion; columns are computed in batches (pure gemm) via one `forward_from(ℓ)` per batch. On a causal decoder (§5.5) the identical construction reads the last position $p = h_{L,\text{last}}$ (the next-token driver) rather than the mean, giving the causal analog. The embedding-space projection (§4.3) removes the component along the unit output $e = p/\|p\|$, since a normalized embedding is invariant to radial scaling: $J_{\text{emb}} = \frac{1}{\|p\|}(I - ee^\top)J$.

Appendix B — Per-layer tables

Structural signatures and internal identity accuracy by residual depth, straight from the ledger.

Authors (MiniLM, 6 layers). Internal identity accuracy by layer: [5.2, 7.2, 8.8, 8.0, 7.2, 8.8]% (chance 1.8%).

layer	stable rank	eff dim	verbaliz.	autocorr
0	22.1	73.5	0.50	-0.004
1	31.4	88.8	0.51	+0.004
2	31.2	93.0	0.53	+0.048
3	37.2	104.6	0.50	+0.101
4	54.5	127.6	0.49	+0.116
5	106.9	205.2	0.61	+0.088

Coders (JinaBERT, 12 layers, 15 developers). Internal identity accuracy by layer: [45.2, 44.8, 48.0, 50.8, 40.0, 48.8, 48.0, 40.4, 42.0, 42.0, 42.4, 43.2]% (chance 6.7%).

layer	stable rank	eff dim	verbaliz.	autocorr
0	30.6	79.6	0.19	-0.033
1	36.7	95.2	0.20	-0.030
2	38.6	98.9	0.21	-0.001
3	35.6	100.7	0.21	+0.023
4	33.4	102.4	0.21	+0.032
5	20.3	91.2	0.21	+0.039
6	37.9	113.8	0.19	+0.062
7	54.3	126.4	0.20	+0.071
8	59.5	132.0	0.20	+0.086
9	61.7	139.0	0.22	+0.085
10	73.3	151.3	0.23	+0.088
11	79.0	185.0	0.22	+0.075

Decoder (GPT-2, 12 layers). Next-token accuracy by residual depth: [0.0, 2.6, 2.6, 3.2, 3.2, 4.7, 5.1, 7.2, 10.9, 13.0, 17.0, 21.3, 22.6]%.

layer	stable rank	eff dim	verbaliz.	autocorr
0	2.6	5.8	0.89	+0.007
1	5.0	16.2	1.81	+0.050

layer	stable rank	eff dim	verbaliz.	autocorr
2	6.3	25.8	2.44	+0.031
3	6.4	30.3	2.08	+0.047
4	6.6	32.1	2.15	+0.044
5	6.7	33.5	1.94	+0.052
6	6.5	31.1	1.90	+0.058
7	6.5	29.9	1.57	+0.099
8	6.7	32.5	1.26	+0.123
9	5.2	23.7	1.39	+0.145
10	3.8	13.8	1.10	+0.161
11	2.0	4.1	0.89	+0.110

Appendix C — Axis definitions, rosters, and OOD probe sets

Prose style axes (`author_axes`): `male↔female` (gender difference-of-means); and the top-2 one-vs-rest classes of the `educated` and `raised` fields: `educated=England↔rest`, `educated=US↔rest`, `raised=England↔rest`, `raised=US↔rest`. Code style axes (`dataset_axes`): `Systems↔rest`, `Scripting↔rest` (the `paradigm` trait). Each axis is the L2-normalized difference of class-mean reference embeddings, a unit Fisher direction, used for both readout and steering.

Rosters. 55 public-domain prose authors (`corpus/authors.toml`, labelled with gender / birth / raised / educated / college) and 15 open-source developers (`corpus/coders.toml`, git-attributed and name-scrubbed, labelled with a `paradigm` = Systems/Scripting trait).

OOD probes for the fingerprint detector (§5.4). Prose (bar 0.30): four held-out author samples plus `source code`, `modern chat`, `biology abstract`, `legalese`. Code (bar 0.51): four held-out coder samples plus `Victorian prose`, `modern chat`, `news headline`, `recipe`.

Appendix D — Audit ledger

Every number above is generated by `jlens/paper/build_ledger.py` into `jlens/paper/ledger.json` (value → source asset + JSON path) and cross-checked by the figure pipeline. Method claims resolve via `jlens/paper/method_map.md`.

References

Alain, Guillaume, and Yoshua Bengio. 2017. "Understanding Intermediate Layers Using Linear Classifier Probes." *ICLR Workshop*.

Anthropic. 2026. *Verbalizable Representations Form a Global Workspace in Language Models*. https://transformer-circuits.pub/2026/workspace.

Arditi, Andy, Oscar Obeso, Aaquib Syed, et al. 2024. "Refusal in Language Models Is Mediated by a Single Direction." *NeurIPS*.

Baars, Bernard J. 1988. *A Cognitive Theory of Consciousness*. Cambridge University Press.

Belrose, Nora, Zach Furman, Logan Smith, et al. 2023. "Eliciting Latent Predictions from Transformers with the Tuned Lens." https://arxiv.org/abs/2303.08112.

Bricken, Trenton, Adly Templeton, Joshua Batson, et al. 2023. "Towards Monosemanticity: Decomposing Language Models with Dictionary Learning." *Transformer Circuits Thread*.

Caliskan-Islam, Aylin, Richard Harang, Andrew Liu, et al. 2015. "De-Anonymizing Programmers via Code Stylometry." *24th USENIX Security Symposium (USENIX Security 15)*.

Dehaene, Stanislas, and Jean-Pierre Changeux. 2011. "Experimental and Theoretical Approaches to Conscious Processing." *Neuron* 70 (2).

Günther, Michael, Isabelle Mohr, Bo Wang, Han Xiao, et al. 2023. *Jina Embeddings 2: 8192-Token General-Purpose Text Embeddings for Long Documents*. arXiv:2310.19923.

Hewitt, John, and Percy Liang. 2019. "Designing and Interpreting Probes with Control Tasks." *EMNLP*.

Hewitt, John, and Christopher D. Manning. 2019. "A Structural Probe for Finding Syntax in Word Representations." *NAACL*.

Hovy, Dirk, and Shannon L. Spruit. 2016. "The Social Impact of Natural Language Processing." *ACL*.

Kornblith, Simon, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. "Similarity of Neural Network Representations Revisited." *ICML*.

Mairesse, François, Marilyn A. Walker, Matthias R. Mehl, and Roger K. Moore. 2007. "Using Linguistic Cues for the Automatic Recognition of Personality in Conversation and Text." *Journal of Artificial Intelligence Research* 30: 457–500.

Mosteller, Frederick, and David L. Wallace. 1964. *Inference and Disputed Authorship: The Federalist*. Addison-Wesley.

Narayanan, Arvind, Hristo Paskov, Neil Zhenqiang Gong, et al. 2012. "On the Feasibility of Internet-Scale Author Identification." *IEEE Symposium on Security and Privacy*.

nostalgebraist. 2020. *Interpreting GPT: The Logit Lens*. https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

Pennebaker, James W., and Laura A. King. 1999. "Linguistic Styles: Language Use as an Individual Difference." *Journal of Personality and Social Psychology* 77 (6): 1296–312.

Radford, Alec, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. *Language Models Are Unsupervised Multitask Learners*.

Ravfogel, Shauli, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. 2020. "Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection." *ACL*.

Reimers, Nils, and Iryna Gurevych. 2019. "Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks." *EMNLP*.

Rimsky, Nina, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2024. "Steering Llama 2 via Contrastive Activation Addition." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Rivera-Soto, Rafael A., Olivia Elizabeth Miano, Juanita Ordonez, et al. 2021. "Learning Universal Authorship Representations." *EMNLP*.

Schler, Jonathan, Moshe Koppel, Shlomo Argamon, and James W. Pennebaker. 2006. "Effects of Age and Gender on Blogging." *AAAI Spring Symposium: Computational Approaches to Analyzing Weblogs*, 199–205.

Schwartz, H. Andrew, Johannes C. Eichstaedt, Margaret L. Kern, et al. 2013. "Personality, Gender, and Age in the Language of Social Media: The Open-Vocabulary Approach." *PLoS ONE* 8 (9): e73791.

Stamatatos, Efstathios. 2009. "A Survey of Modern Authorship Attribution Methods." *Journal of the American Society for Information Science and Technology* 60 (3): 538–56.

Turner, Alexander Matt, Lisa Thiergart, Gavin Leech, et al. 2023. *Activation Addition: Steering Language Models Without Optimization*. https://arxiv.org/abs/2308.10248.

Vig, Jesse, Sebastian Gehrmann, Yonatan Belinkov, et al. 2020. "Investigating Gender Bias in Language Models Using Causal Mediation Analysis." *NeurIPS*.

Wang, Wenhui, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers." *NeurIPS*.

Wegmann, Anna, Marijn Schraagen, and Dong Nguyen. 2022. "Same Author or Just Same Topic? Towards Content-Independent Style Representations." *Proceedings of the 7th Workshop on Representation Learning for NLP (RepL4NLP)*.

Zou, Andy, Long Phan, Sarah Chen, et al. 2023. *Representation Engineering: A Top-down Approach to AI Transparency*. https://arxiv.org/abs/2310.01405.